Review

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Mammaprint™: a comprehensive review

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The number of breast cancer (BC) cases is growing worldwide, being most frequently diagnosed in the early-setting. Mammaprint™ is a 70-gene-expression signature, originally designed for selecting early BC patients with low risk of developing metastasis, so that they could be spared adjuvant chemotherapy. Its use as a prognostic biomarker has been extensively validated, both retrospectively and prospectively. However, its value as a predictive tool and as a clinically useful tool remains controversial. This review will describe how the test works, its application in the clinic and its limitations. Cost–effectiveness studies will be summarized. Finally, we will provide a perspective on the use of Mammaprint in the near future, as a valuable tool for personalizing the treatment of early BC patients.

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Breast cancer (BC) is diagnosed every year in approximately 1.2 million women worldwide, and is, in industrialized societies, most often localized (early BC) and curable [1]. Curative treatment of BC is based on surgical removal of the primary tumor, commonly followed by radiotherapy and, for tumors considered to be at high risk of relapse, by adjuvant chemotherapy [2]. Advances in adjuvant chemotherapy have significantly improved outcomes for women with early BC [3], but at the cost of important short- and long-term consequences for both patients and society [4,5]. Therefore, in the last 15 years, an intense debate has developed on the most appropriate means of defining which patients are at a sufficiently high risk of relapse to justify the use of chemotherapy [2].

Traditionally, a host of tumor and patient characteristics are taken into consideration to determine whether a patient is at high risk of relapse. These essentially phenotypical tumor characteristics are used to infer disease behavior. However, disease behavior is also driven by alterations occurring at the molecular level [6]. Advances in genome sequencing in the 2000s have led to multiple initiatives to go beyond disease phenotype into the genomic, transcriptomic and proteomic tumor landscapes, ultimately revealing the existence, under the overarching umbrella of BC, of multiple distinct pathologic entities [7]. This new molecular way to conceptualize BC, as well as the self-evident limitations of traditional risk assessment methods, gave birth to a number of attempts to create tools capable of providing a glimpse into disease biology to aid decision making – the gene-expression signatures (GES) [8].

Today, a number of GES measured on primary tumor tissue have been developed and are in current clinical use, holding the possibility of sparing a substantial proportion of patients not only from overtreatment but also from undertreatment. However, the expense related to performing GES and the absence, until recently, of prospective validation of their clinical utility have continued to fuel the debate surrounding their use.

Mammaprint™ was the first GES approved for use by the US FDA and was the first one to have its clinical utility validated prospectively as decision-making tool when results are discordant with classic criteria in a prospective randomized trial (MINDACT) [9,10]. This article will provide a thorough and critical review of Mammaprint, including its preclinical development, the studies which form the basis of its use in the clinic as well as how it compares to other available GES and its future prospects.

How was the signature developed?

Mammaprint was initially developed in archived fresh-frozen samples from a set of 78 young BC patients with tumor size less than 5 cm and negative lymph nodes (LN), not treated with adjuvant systemic therapy (Supplementary Table 1 & Figure 1). In this group, around half remained free of distant metastasis at five years [9]. The aim

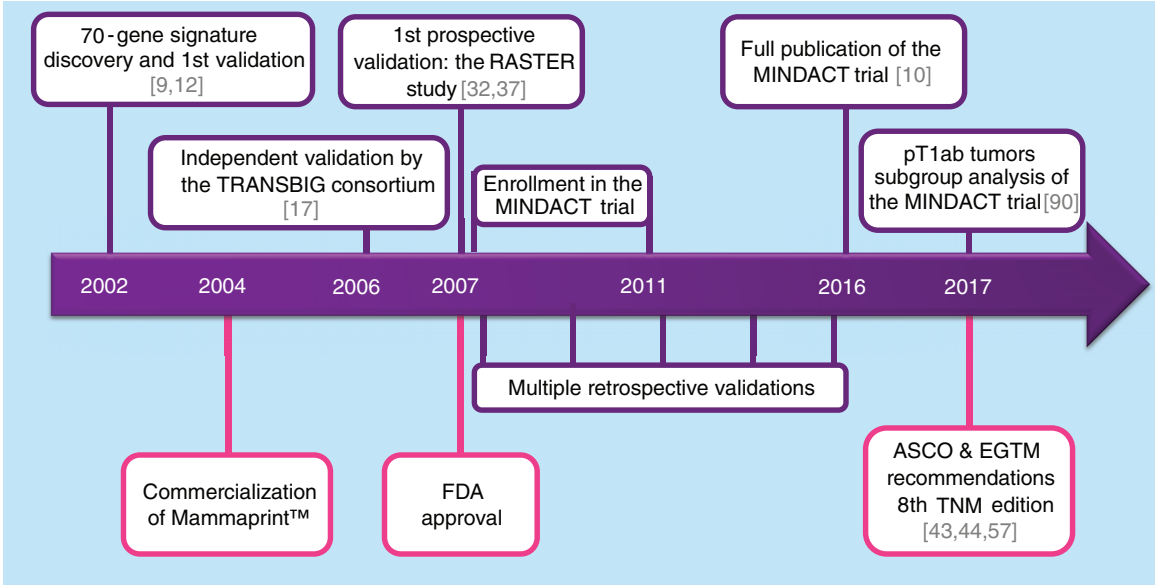

Figure 1. Clinical development of Mammaprint™, from 2002 to 2017.
ASCO: American Society of Clinical Oncology; EGTM: European Group on Tumor Marker; TNM: American Joint Committee on Cancer 8th Edition Cancer Staging Manual – Breast Cancer.

of the researchers was to develop a gene-signature that could differentiate the subgroup of patients with a low risk for distant metastasis from those experiencing metastasis in the first years after surgery. From the initial approximately 5000 genes tested by supervised analysis, a signature was created with 70 genes that best correlated with prognosis. After adjusting for classical prognostic factors, a multivariate model showed that the signature was a strong prognostic factor: patients with a 'poor-prognosis signature' had an odds ratio (OR) of 18 (95% CI: 3.3–94) of developing distant metastasis within five years compared with the ones with the 'good-prognosis' signature. The 70 genes included pertain predominantly to one of the six classical hallmarks of cancer (Figure 2) [6], largely covering proliferation, resistance to apoptosis, invasion and dissemination.

Today, the signature is marketed by Agendia under the name of Mammaprint. The test was originally developed on fresh-frozen tissue, which hampered its practical use by oncologists as well as its validation on biobanked tumor samples of clinical trials with a long follow-up, but today it can be performed in formalin-fixed paraffin-embedded tissue, with equivalent analytical validity [35].

Historical perspective on the development of Mammaprint

Early validation studies

Following initial development, most efforts consisted in retrospective validation series and pooled analysis, assessing the value of the gene-signature in different patient populations and settings [11–20,28–34] (Supplementary Table 1 & Figure 1). Patients who had neoadjuvant therapy were excluded from all these studies and as 'distant metastasis-free survival' (DMFS) was the end point used to develop the 70-gene signature, it remained the main end point of most studies. Taken together, the results of all these studies shed light not only on the potential of the test but also on important aspects of BC biology.

High prognostic value of Mammaprint for predicting distant relapse

It has long been recognized that BC has the ability to metastasize to distant sites even at an early point of tumor development. Even in small tumors (≤2 cm), Mammaprint can predict which patients have a higher risk of developing distant metastasis: 28% of 'high-genomic risk' patients by the gene-signature had a distant relapse at 10 years, compared with 13% of the 'low-genomic risk' patients (adjusted hazard ratio [HR]: 2.43; 95% CI: 1.6–3.8) [30]. This effect remained even in the very small pT1ab tumors (HR: 3.45; 95% CI: 1.0–11.5), thus further confirming the ability of Mammaprint to capture this intrinsic metastatic potential.

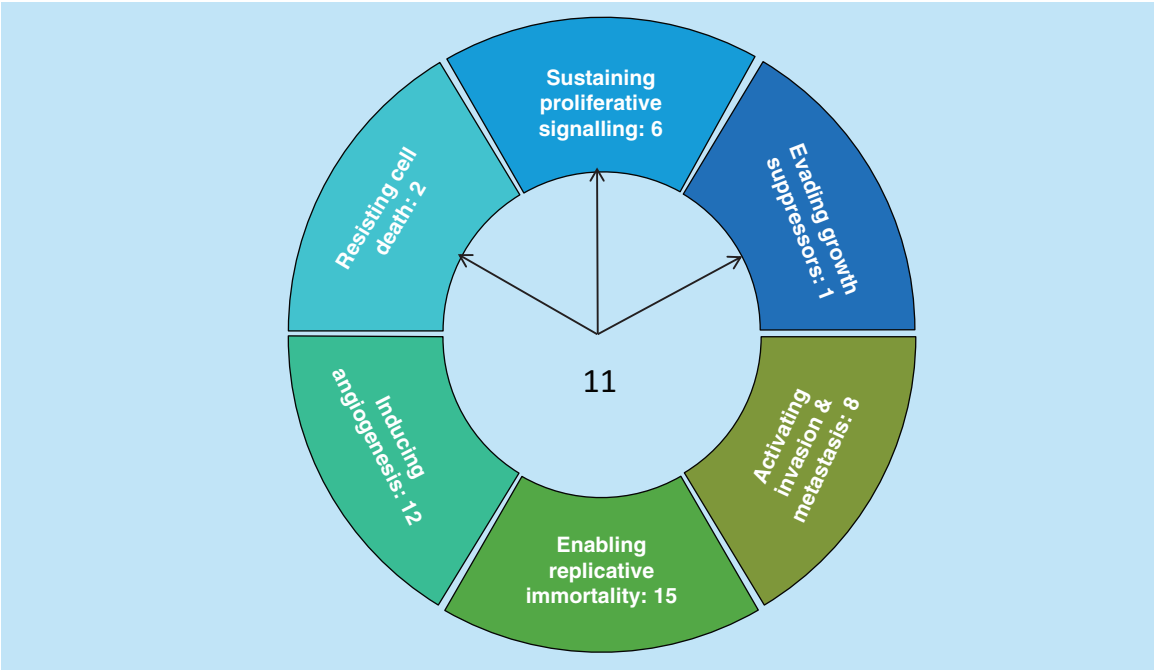

Figure 2. Hallmarks of cancer and number of genes in Mammaprint™. Mammaprint is composed by 70 genes, most of them associated with one or more hallmark. Sustaining proliferative signaling: *FLT1, HRASLS, STK32B, RASSF7, DCK, MELK, EXT1, GNAZ, EBF4, MTDH, PITRM1, QSCN6L1*; Evading growth suppressors: *TGFB3, FLT1, HRASLS, STK32B, RASSF7, DCK, MELK, EXT1, GNAZ, EBF4, MTDH, PITRM1, QSCN6L1; Activating invasion and metastasis: COL4A2, GPR180, MMP9, GPR126, RTN4RL1, DIAPH3, CDC42BPA, PALM2*; Enabling replicative immortality: *CCNE2, ECT2, CENPA, LIN9, KNTC2, MCM6, NUSAP1, ORC6L, TSPYL5, RUNDC1, PRC1, RFC4, RECQL5, CDCA7, DIL*; Inducing angiogenesis: *ALDH4A1, AYTL2, OXCT1, PEC1, GMPS, GSTM3, SLC2A3, FLT1, FGF18, COL4A2, GPR180, EGLN1, MMP9*; Resisting cell death: *BBC3, EGLN1, FLT1, HRASLS, STK32B, RASSF7, DCK, MELK, EXT1, GNAZ, EBF4, MTDH, PITRM1, QSCN6L1*. 11 genes pertain to three hallmarks at the same time (indicated by arrows). 16 genes relate to other or unknown functions: *LOC100288906, C9orf30, ZNF533, C16orf61, SERF1A, C20orf46, LOC730018, LOC100131053, AA555029_RC, LGP2, NMU, UCHL5, JHDM1D, AP2B1, MS4A7, RAB6B*.

LN status & value of Mammaprint

LN status [11] and, in particular, the number of positive LN [20] were not correlated with a gene-signature category, indicating that dissemination to distant sites (hematogenous spread) is distinct from dissemination to LN (lymphogenic spread). LN-positive patients (between one and three positive nodes) with a low-genomic risk had a lower risk of distant metastasis (DMFS of 98% at five years and 91% at ten years), compared with high-genomic risk patients (DMFS of 80% at 5 years and 76% at 10 years) [14]. These survival data are derived from patients treated between 1984 and 1996, of whom only 18% received endocrine treatment (ET) and none received chemotherapy; therefore, the DMFS rate of low-genomic risk patients was considered favorable, indicating that this patient subgroup could possibly also be spared chemotherapy [11,12,14]. However, in patients with four to nine positive LN [20], the low-genomic risk group had a DMFS at five years of 87% and at ten years of 67%, which do not justify withholding chemotherapy, demonstrating that tumor burden is still an important prognostic factor [14].

Combination with clinical risk indexes

Clinical risk indexes like the St. Gallen [36], Adjuvant! Online (AOL) [37,38] and others, are commonly used in clinical practice to predict DMFS or overall survival (OS) and, consequently, to select patients for adjuvant therapy. However, when adjusting these clinical risk indexes for the gene-signature, most of them did not retain an independent association with DMFS or OS [13], hinting that classic clinical risk factors are mainly a reflection of the intrinsic genetic properties of the tumor. Still, when combining clinical risk factors with the gene-signature in a multivariate model for the detection of distant metastasis, the area under the curve (AUC) value increased from 0.70 (95% CI: 0.61–0.79) to 0.76 (95% CI: 0.68–0.85) and the AUC for OS increased from 0.74 (95% CI: 0.66–0.82) to 0.82 (95% CI: 0.75–0.89) [13]. Furthermore, multiple studies have demonstrated that patients

classified as having a high-clinical risk but a low-genomic risk had a similar survival compared with patients with low-clinical/low-genomic risk [15,17,22,31]. Moreover, Mammaprint presented a very high negative-predictive value for the development of distant metastasis at 5 years (over 85%) [16,18], thus confirming that a patient presenting with a low-genomic risk has a very low probability of developing distant metastasis as an early event. In Supplementary Table 1, there is a description of the rate of discordance between the gene-signature and the clinical-risk assessment in most studies, specifying in brackets the percentage of patients classified as high-clinical risk/low-genomic risk that could eventually avoid chemotherapy.

Postmenopausal patients

Is Mammaprint, initially developed in young BC patients, also useful in older, exclusively postmenopausal populations? In one series [16], only 27% of patients had a low-genomic risk, and 10-year DMFS was good both in low- and high-genomic risk groups: 100 versus 90%, respectively. However, in another series [17], 61% of patients had a low-genomic risk and it was demonstrated that Mammaprint was strongly prognostic; low-genomic risk patients had a 5-year DMFS of 93% compared with 72% in high-genomic risk patients (HR: 4.6; 95% CI: 1.8–12.0). This was further confirmed in pooled analyses in which more than half of the patients were postmenopausal [28,33,34].

Prognostic ability over time

In the updated analysis of the first validation series, after a median follow-up of 18.5 years, the HR for DMFS was 9.6 (95% CI: 4.2–22.1) in the first 5 years, but it was only 1.1 (95% CI: 0.5–2.5) in years 5–10 and 1.2 (95% CI: 0.2–6.0) in years 10–15 [12]. A similar trend was observed for OS. Therefore, the prognostic ability of Mammaprint may be reduced over time, being higher in the first five years after diagnosis and then declining. Nonetheless, the low number of long-term events could have contributed to these numbers. As the EBCTCG meta-analysis [3] has shown that the beneficial effect of chemotherapy mostly happens in the first five years after diagnosis, this does not diminish its value as an adjuvant treatment decision making tool. However, at least one study suggests that Mammaprint can identify an ultralow-risk population (score $\geq$0.355) with a breast cancer-specific survival (BCSS) rate between 94 and 97% at 20 years [21]. A meta-analysis showed that proliferation is the main component of the prognostic performance of Mammaprint and other GES [39] – in an analysis of the TransATAC study, GES based on cell proliferation showed a good performance for the prediction of late relapses, thus one can speculate that Mammaprint may have the same ability [40].

The RASTER trial

All of the aforementioned studies were retrospective and based on data from patients with different follow-up times, receiving a variable amount of adjuvant therapy, who are not representative of the modern BC patient populations, in which almost all women with estrogen receptor (ER)-positive tumors receive ET [41]. Therefore, a prospective study was still required to validate Mammaprint, but some logistical problems needed to be addressed first. The RASTER study [23] evaluated the feasibility of performing the gene-signature using fresh-frozen tissue in community hospitals in The Netherlands as well as how it changed physicians' prescription of systemic therapy. Among evaluable patients, almost half had high-genomic risk, but only 43% had high-clinical risk according to the Dutch guidelines [42]. Interestingly, due to the fact that physicians' prescription of systemic treatment was based on the Dutch guidelines, the use of Mammaprint actually resulted in an increase in the number of patients receiving chemotherapy and ET. Nevertheless, the RASTER study proved that Mammaprint is a feasible test to use in community hospitals [23]. The updated follow-up shows that among low-genomic risk patients, 5-year DMFS is identical whether or not AOL indicates a low- (94.3%) or a high-clinical risk (97.6%) [22].

The MINDACT trial

The MINDACT trial enrolled 6693 women with up to three positive LN, in nine European countries, with the goal of proving the clinical utility of adding Mammaprint to standard clinical–pathological criteria, for an improved selection of patients for adjuvant chemotherapy [10]. More specifically, the trial aimed to reduce the prescription of adjuvant chemotherapy by using the Mammaprint test, without impairing outcome. Modified AOL was used to determine high- and low-clinical risk (Figure 3) and patients were classified into four groups according to their clinical and genomic-risk: low-clinical/low-genomic, high-clinical/high-genomic, high-clinical/low-genomic and low-clinical/high-genomic. Randomization between chemotherapy versus no-chemotherapy only occurred in the discordant groups (high-clinical/low-genomic and low-clinical/high-genomic). The main end point was to prove

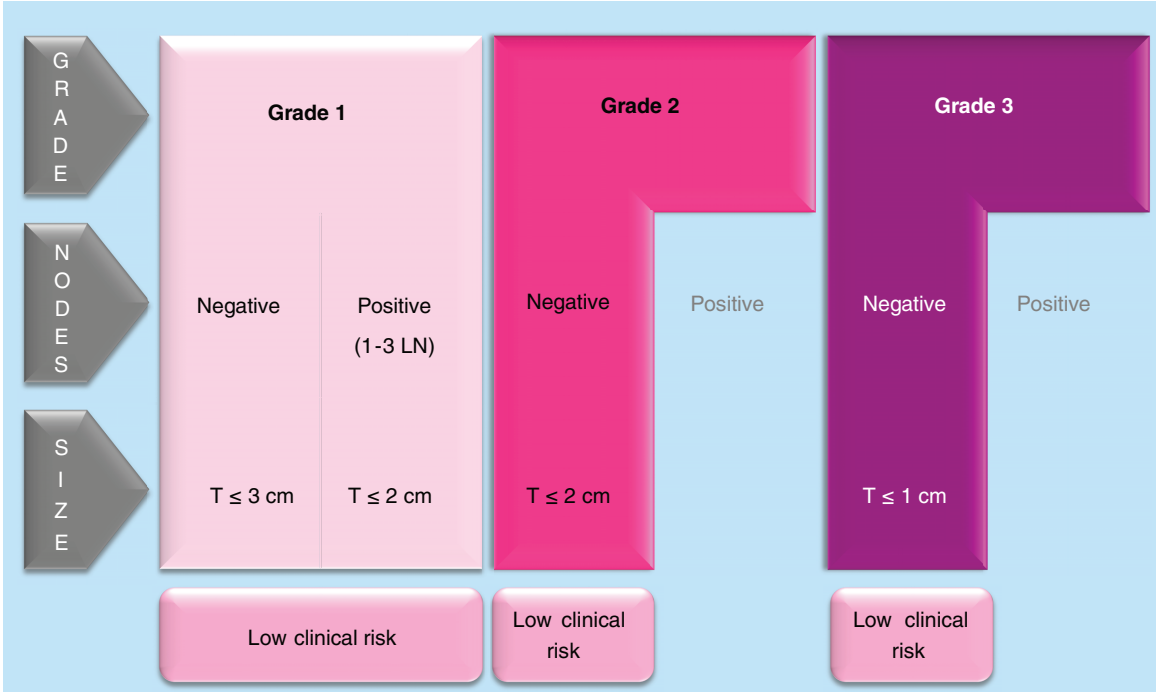

Figure 3. Definition of 'low-clinical risk' in the MINDACT trial for estrogen-receptor positive tumors.
LN: Lymph node; T: Breast tumor size.

that among the cohort of patients with high-clinical/low-genomic risk who did not receive chemotherapy, the lower boundary of the 95% CI for the rate of 5-year DMFS would be 92% or higher. As such, the MINDACT trial should be viewed as a single-arm 'non-inferiority' study, as it was clearly not powered to perform direct statistical comparisons between high-clinical/low-genomic risk patients who received versus who did not receive chemotherapy (these comparisons were secondary end points).

The results have shown that the primary end point was met: the 5-year DMFS in the primary-test population (high-clinical/low-genomic risk patients assigned not to receive chemotherapy and excluding the ones with a change in risk) was 94.7% with a 95% CI ranging from 92.5 to 96.2%, thus not crossing the noninferiority lower boundary (Table 1). Additionally, in the high-clinical/low-genomic intention-to-treat (ITT) population, patients who received chemotherapy had a 5-year DMFS rate of 95.9%, whereas those without chemotherapy had a 5-year DMFS rate of 94.4% – this absolute difference of 1.5% corresponded to a nonsignificant HR of 0.78 (95% CI: 0.50–1.21). As the study was not powered to make a direct comparison of outcomes between these groups, it is not strictly possible to state that their 5-year DMFS are not different. However, the magnitude of difference (1.5%) may not be viewed by many patients as valuable enough to justify the chemotherapy-associated side effects and long-term risks. On the other hand, patients with low-clinical/high-genomic risk had very similar 5-year DMFS rates whether they received chemotherapy or not (95.8 vs 95.0%, respectively), with a HR of 1.17 (95% CI: 0.59–2.28). Therefore, in patients with low-clinical risk, the clinical utility of using Mammaprint could not be demonstrated.

The authors calculated that among the high-clinical risk group, the use of Mammaprint would lead to a reduction in the use of chemotherapy in 46% of these patients (the ones with low-genomic risk), corresponding to 23% of the total population. Finally, the authors assessed the rate of 5-year DMFS if only clinical-risk or genomic-risk scores would be applied to define which patients should receive chemotherapy and the results were strikingly similar: 95.0% with clinical-risk evaluation and 94.7% with genomic-risk assessment. Thus, the added value of Mammaprint is that with its use as an aid to treatment selection, the 5-year DMFS is maintained, but an additional 23% of the overall population will not receive chemotherapy, which may decrease toxicities and costs.

Regarding 5-year disease-free survival (DFS) and OS, there were no differences between the chemotherapy versus no-chemotherapy groups within risk-discordant patients in the ITT population. However, in the per-protocol population, 5-year DFS was slightly worse in the high-clinical/low-genomic risk group without chemotherapy

Table 1. Results of the MINDACT prospective trial.

Number of patients and FU duration	Characteristics	5-year DMFS in each subgroup	5-year DMFS HR (95% CI)	5-year OS HR (95% CI)	5-year DFS (95% CI)	5-year DMFI (95% CI)
N = 6693 total High-C[†]/low-G[‡]: 23% Low-C[†]/high-G[‡]: 9% Low-C[†]/low-G[‡]: 41% High-C[†]/high-G[‡]: 27% Primary-test population: n = 644 ITT population[¶]: n = 2207 PPS population[¶]: n = 1706 Median FU: 5.0 years	≤70 years (67% were ≥50 years; median 55 years); pT1–3 N0–1 (21% N1); 88% ER^{+}[§] and/or PgR^{+}[§]; 10% HER2^{+} Hormone-R^{+}/HER2^{-}: 81% HER2^{+}/Hormone-R^{+}: 8% HER2^{+}/Hormone-R^{-}: 2% TNBC: 10%	Low-C/low-G: 97.6% High-C/low-G: 95.1% Low-C/high-G: 94.8% High-C/high-G: 90.6%	Primary-test population (high-C/low-G, no CT): 94.7% (92.5–96.2). High-C/low-G ITT population: with CT: 95.9% (94.0–97.2) vs no CT: 94.4% (92.3–95.9) – Δ 1.5%. HR: 0.78 (0.5–1.2) High-C/low-G PPS population: with CT: 96.7% (94.7–98.0) vs no CT: 94.8% (92.6–96.3) – Δ 1.9%. HR: 0.65 (0.38–1.10)	High-C/low-G ITT population: with CT: 98.4% (97.0–99.1) vs no CT: 97.0% (95.4–98.1) – Δ 1.4%. HR: 0.69 (0.4–1.4) High-C/low-G PPS population: with CT: 98.8% (97.4–99.5) vs no CT: 97.3% (95.6–98.4) – Δ 1.5%. HR: 0.63 (0.29–1.37)	High-C/low-G ITT population: with CT: 92.9% (90.5–94.7) vs no CT: 90.1% (87.5–92.1) – Δ 2.8%. HR: 0.71 (0.5–1.01) High-C/low-G PPS population: with CT: 93.3% (90.7–95.2) vs no CT: 90.3% (87.6–92.4) – Δ 3%. HR: 0.64 (0.43–0.95)	High-C/low-G ITT population: with CT: 96.6% (94.8–97.8) vs no CT: 95.3% (93.4–96.6) – Δ 1.3%. HR: 0.76 (0.5–1.2)
High-C[†]/low-G[‡] risk, only N0 patients (n = 812)			High-C/low-G ITT population: with CT: 95.7% (93.0–97.4) vs no CT: 93.2% (90.1–95.4) – Δ 2.5% ER^{+}/HER2^{-} patients in the ITT population (n = 676): with CT: 95.5% (92.5–97.3) vs no CT: 93.9% (90.6–96.1) – Δ 1.6%. HR: 0.80 (0.4–1.5)	NR	NR	NR
Only pT1abN0 patients (n = 826) [43] High-C[†]/low-G[‡]: 0.6% Low-C[†]/high-G[‡]: 24% Low-C[†]/low-G[‡]: 76% High-C[†]/high-G[‡]: 0%	≤70 years (38% were ≥60 years), 64% postmenopausal; 91% ER^{+}; 6% HER2^{+}; Hormone-R^{+}/HER2^{-}: 85% (59% Luminal A; 26.5% Luminal B) HER2^{+}/Hormone-R^{+}: 5% HER2^{+}/Hormone-R^{-}: 1% TNBC: 5%	NR	Low-C/high-G with CT: 97.3% (89.4–99.3) vs no CT: 91.4% (82.6–95.9) – Δ 5.9%	Low-C/high-G with CT: 98.5% (89.6–99.8) vs no CT: 95.8% (89.1–98.4) – Δ 2.7%	Low-C/high-G with CT: 92.3% (83.5–96.5) vs no CT: 84.5% (74.5–90.8) – Δ 7.8%	Low-C/high-G with CT: 98.8% (91.9–99.8) vs no CT: 91.4% (82.6–95.9) – Δ 7.4%

[†]Clinical risk group: as defined by the Adjuvant! Online (version 8.0 with HER2 status) – patients were considered as having high-clinical risk when the 10-year breast cancer-specific survival was predicted to be less than 88% for ER-positive tumors, and less than 92% for ER-negative tumors, respectively.
[‡]Genomic risk group: as defined by the 70-gene-expression signature (Mammaprint™)
[§]Hormone receptor positivity was defined as the presence of at least 1% of immunoreactive cells, an Allred score of more than 2 (on a scale from 0 to 8, with higher scores indicating a greater number of receptors), or a level of cytosolic protein of at least 10 fmol per milligram.
[¶]Includes both high-clinical/low-genomic risk patients and low-clinical/high-genomic risk patients.
Prospective trial: MINDACT [10] (2016). Hazard ratios compare the chemotherapy group versus the no-chemotherapy group (which is the reference).
CT: Chemotherapy; DFS: Disease-free survival (time from surgery to any recurrence [local or regional], distant metastasis, ductal carcinoma *in situ*, second breast primary, invasive second primary cancer or death from any cause); DMFI: Distant metastasis-free interval (time from surgery until the diagnosis of a distant recurrence); DMFS: Distant metastasis-free interval (time from surgery until the diagnosis of a distant metastasis or death by any cause); ER: Estrogen receptor; FU: Follow-up; Hormone-R: Hormone receptor; HR: Hazard ratio; ITT: Intention-to-treat; N0: Patients with lymph-node negative tumor; NR: Not reported; OS: Overall survival (time from surgery until death from any cause); PgR: Progesterone receptor; PPS: Per-protocol population; TNBC: Triple-negative breast cancer.

compared with patients who received chemotherapy: 90.3 versus 93.3%, respectively (HR: 0.64; 95% CI: 0.43–0.95; p = 0.03). Additionally, among patients with ER-positive/HER2-negative/LN-negative disease, the 5-year DMFS was similar between the chemotherapy and no-chemotherapy groups: 95.5 versus 93.9%, respectively (HR: 0.80; 95% CI: 0.44–1.45).

Problems & controversies regarding the test & the MINDACT trial

Technical challenges

There are several technical challenges regarding the use of Mammaprint. As proliferation rate is a continuous variable, there must be a trade-off in each cut-point used between the amount of patients spared of adjuvant systemic treatment and the proportion who may relapse in the future. It must be emphasized that the cut-point used for risk-classification applies to western populations and usually splits the population in half in terms of

risk (Supplementary Table 1); this cut-point is probably inadequate for Asian populations, as more than 80% are classified as high-genomic risk [18,19]. Furthermore, other cut-points can be used to classify patients as ultralow-risk [21] or as ultrahigh-risk [44,45], but these have been less extensively studied.

In the MINDACT trial, 162 patients were falsely identified as being high-genomic risk due to a change in RNA-extraction solution, showing that technical issues can interfere with the reliability of the test [10].

MINDACT design

Definition of the primary end point

A major criticism to the MINDACT trial is that the primary end point is based in a nonrandomized cohort of patients: high-clinical risk/low-genomic risk patients not receiving chemotherapy. Besides, a formal noninferiority design was not pursued, as it would lead to a much higher sample size, which was not feasible in terms of cost and time. However, treatment de-escalation in oncology would be impossible if we would always require a 'non-inferiority' design, due to its associated high costs and low interest from commercial companies to fund it. Therefore, other de-escalation studies have used this nonrandomized approach and their results were welcomed by both physicians and patients. An example is the single-arm APT trial, which tested the use of chemotherapy without anthracyclines plus trastuzumab for one year in 410 patients with small (≤2 cm) HER2-positive tumors [46]. Invasive DFS rate at only three years was 98.7% (95% CI: 97.6–99.8) and this leads to a practice-changing de-escalation strategy in these patients, sparing them from the long-term effects of anthracyclines. Updated 7-year results confirmed the initial favorable prognosis [47]. Therefore, this regimen was endorsed by scientific guidelines, without the need for evaluating the potential harm associated with anthracyclines removal in a randomized fashion. Likewise, Mammaprint was also endorsed by scientific societies [48,49], even if the secondary end point shows that there might be a potential small gain in DMFS/DFS with the prescription of chemotherapy, as it was recognized that this small gain would be offset by the chemotherapy-associated side effects and risks.

Definition of clinical risk

The TRANSBIG consortium chose AOL to define clinical risk, as it was a validated, standardized assessment, freely available to any clinician and based on large clinical databases [50]. The TRANSBIG consortium, with the involvement of EUROPA DONNA, agreed that a 92% chance of survival at ten years without chemotherapy was a desirable goal, as any benefit from chemotherapy beyond this threshold would be of small magnitude and counter-balanced by its risks (namely premature menopause, cognitive dysfunction, cardiac damage and leukemia [4,51]). As adjuvant ET for women with ER-positive disease provides an estimated 4% survival gain at ten years, the threshold for this group was set at 88%.

However, AOL did not include the levels of ER, PgR and Ki67 positivity to stratify Luminal A versus Luminal B tumors. Additionally, it might no longer be accessible [38]: however, the statisticians of MINDACT have provided a table that defines clinical risk as obtained with the use of AOL during the conduct of MINDACT [10].

Definition of the DMFS lower boundary

In noninferiority trials, the definition of a threshold for what is considered 'not clinically significant' is always a matter of subjectivity and debate. Therefore, when the MINDACT trial was designed, a large discussion was set within the TRBANSBIG consortium to define the 5-year DMFS lower boundary for MINDACT [52]. As explained above, it was decided that a 5-year DMFS ≥92% would be considered acceptable. However, this definition should be regarded as a trade-off in terms of the proportion of patients who have a distant relapse versus those who are safely spared from chemotherapy. Therefore, this compromise cannot be considered as an 'applies-to-all' decision and should be explained to the patient before ordering Mammaprint.

Potential chemotherapy benefit in 5-year survival rates

The MINDACT trial was not powered to show differences in survival according to the prescription of chemotherapy. However, it is true that in the high-clinical risk/low-genomic risk ITT and per-protocol populations, the HR for 5-year DMFS, OS and DFS was always below 1.0, favoring chemotherapy, including in ER-positive/HER2-negative LN-negative patients (Table 1). A reason may be that 8% of these patients had HER2-positive tumors and, probably, they were undertreated with ET alone. Another explanation for this possible positive effect of chemotherapy in low-proliferative tumors may reside in its indirect endocrine effects: 35% of the high-clinical/low-genomic risk patients were ≤50 years, likely being premenopausal, and 98% were hormone receptor-positive. Thus, chemotherapy

might have induced at least a temporary ovarian function suppression in two-thirds of these young patients and, as the SOFT trial has shown, ovarian suppression can lead to small improvements in 5-year DFS and DMFS when added to tamoxifen, but OS gain was only seen in the high-risk women who also got chemotherapy [53]. The TAILORx trial randomized patients with ER-positive/HER2-negative LN-negative tumors and an Oncotype DX® intermediate recurrence score (11–25) to ET versus chemotherapy + ET: in the ITT population, there were no survival differences between the two groups; however, in the subgroup of women ≤50 years and with a recurrence score of 16–25, there was an invasive DFS benefit with chemotherapy + ET versus ET alone, which may be attributed to a premature menopause induced by chemotherapy. Yet, it is important to keep in mind that these findings come from a nonpreplanned subgroup analysis and should not be considered as definitive [54].

Long-term survival

The recently published European Network for Health Technology Assessment report expresses the concern that the possible decrease in DMFS at five years could widen in the next five years and result in a substantial harm in terms of 10-year OS [55]. However, this increase in distant recurrences in the no-chemotherapy arm versus the chemotherapy arm seems highly unlikely: the EBCTCG meta-analysis involving 100,000 patients showed that the effect of chemotherapy is mainly observed during the first five years after treatment and that after this initial period the survival curves do not further separate [3]. Thus, it seems fair to assume that 10-year results from MINDACT will show similar rates of recurrence events in both trial arms. Moreover, it is not possible to statistically 'model' a prediction for the 10-year OS, as the population included in this trial is a new highly selected population defined by a diagnostic test not available in the past.

Additionally, in the overall population, DFS events corresponded mostly to second primary cancers (42%), followed by distant metastases (36%), locoregional recurrences (16%) or deaths due to any cause (6%). As most second primary cancers and locoregional recurrences can still be treated with curative intent, it is unlikely that substantial harm in OS will emerge in the no-chemotherapy arm.

Quality of life

Earlier studies stated that the reduction on the rate of prescription of chemotherapy among the high-clinical/low-genomic risk group could translate into reduced toxicity and thus better quality of life (QoL), but no formal evaluation of QoL was performed in the MINDACT trial [10]. The reason is that it was perceived by both authors and patient advocates that it would be too burdensome for the patients and additionally, at the time, there was no validated QoL instrument [56]. While the utility of using Mammaprint in terms of QoL cannot be stated, common clinical sense would indicate that for the 23% of patients who are spared the side effects of chemotherapy, there is a QoL benefit extending over a few months or even a few years in case of severe cognitive dysfunction, sleep disturbances and fatigue [51,57,58].

Predictive ability

Kok *et al.* [27] evaluated the ability of Mammaprint to predict which patients benefit the most from adjuvant treatment with tamoxifen. However, in these ER-positive patients receiving adjuvant tamoxifen, being 'incomplete-endocrine responsive' (either ER or progesterone receptor [PgR] between 0 and 49%, with at least ER >10%) was more predictive of a worse BCSS (HR: 7.22; 95% CI: 2.17–24.0) than having a high-genomic risk (HR; 2.78; 95% CI: 1.3–5.9). After adjustment for ER/PgR positivity and other clinical factors, Mammaprint was no longer significantly associated with BCSS (HR: 1.8; 95% CI: 0.77–4.61).

The predictive ability of Mammaprint for the benefit of adjuvant chemotherapy has not been prospectively demonstrated, as the MINDACT trial was not powered to directly compare outcomes between chemotherapy versus no-chemotherapy in the clinical/genomic-risk discordant groups [10]. This is a limitation, as the greatest utility of a genomic testing is to guide decisions regarding the use of chemotherapy and, for that, it should be able to predict which patients will benefit from chemotherapy or not. This means that, for now, Mammaprint's ability to select patients for chemotherapy is based solely on its prognostic capabilities.

Use in HER2-positive or TNBC patients

In a retrospective pooled analysis in HER2-positive disease, among 89 patients, all chemotherapy and trastuzumab-naive, 22% were classified as low-genomic risk, which is an intriguing observation as HER2-positive BC is considered an aggressive disease [29]. Low-genomic risk patients had a 10-year DMFS of 84% in the absence of chemotherapy

and trastuzumab, while high-genomic risk patients had a worse outcome (10-year DMFS of 55%). However, in the MINDACT trial, HER2-positive patients constituted only 8% of the high-clinical/low-genomic risk population (n = 124) and a separate analysis was not performed in these patients. Thus, the value of Mammaprint in this population is uncertain [10].

Likewise, the usefulness of the gene-signature is probably very low or absent in triple-negative patients, as 90–96% of them were classified as high-genomic risk in the MINDACT trial [10] and in earlier validation series [20,31,33]. This is in agreement with data from a large meta-analysis of GES, in which most of the HER2-positive and triple-negative tumors were classified as high-risk, therefore limiting its clinical utility in these patients. Another comprehensive meta-analysis using both clinicopathologic and gene-expression data and focusing on the main molecular subtypes showed that among ER-positive/HER2-negative tumors, proliferation-related genes were indeed the strongest prognostic parameter; however, in the triple-negative subset, the most important prognostic genes were the ones related with immune response and in the HER2-positive group, tumor invasion-related genes were the more relevant ones [59].

Use of Mammaprint in clinical trials

Mammaprint is being used in clinical trials, most notably in the I-SPY2 neoadjuvant trial (NCT01042379). Here, trial population is categorized into eight subtypes defined by three biomarkers: hormone receptors (ER/PgR), HER2 and Mammaprint. The gene-signature was chosen as a biomarker, as it was previously shown that a high-genomic risk was associated with higher neoadjuvant chemosensitivity [24]. For this trial, patients traditionally classified as high-genomic risk by Mammaprint were further divided in two categories: high-risk1 (MP1) and (ultra)high-risk2 (MP2) based in a higher cut-point [44]. The MINT I study also demonstrated that patients classified as MP2 had a greater sensitivity to neoadjuvant chemotherapy, as their pCR rate was 44% compared with a pCR rate of 24% in MP1 patients [45].

Practical application in the clinic

Mammaprint was the first GES approved by the FDA, in 2007. Following the publication of MINDACT, both the American Society of Clinical Oncology and the European Group on Tumour Markers published updated guidelines recommending the use of Mammaprint [48,49]. The National Comprehensive Cancer Network still favors the use of Oncotype DX, though it recognizes Mammaprint™ as a valid alternative [41]. Consensus opinion is, today, that a low-genomic risk Mammaprint result can be used in patients with ER-positive/HER2-negative early BC, with up to three positives nodes, considered as having a high-clinical risk of relapse according to traditional criteria, to safely forego the use of adjuvant chemotherapy. Additionally, Mammaprint was incorporated in the 8th TNM edition for the classification of BC, along other GES, as a way of downstaging selected ER-positive/HER2-negative LN-negative tumors with low-genomic risk into the same prognostic category as T1a-T1bN0M0 [60]. It should, nonetheless, be discussed with the patient that there might be a small loss in DMFS and DFS with the avoidance of chemotherapy in this setting. A practical approach to the application of Mammaprint in the clinic for patients with ER-positive/HER2-negative tumors is provided in Figure 4.

As Mammaprint's prognostic performance is driven by genes related to proliferation [39], an open question is if Mammaprint can be replaced by a simpler and cheaper method of evaluating cell proliferation, like nuclear grade or the Ki67 score. A study tested the Genomic Grade (a GES) in 883 tumor samples of patients enrolled in the BIG 1–98 trial and showed that among the 60% of patients who were centrally classified as nuclear grade 2, the ones classified as GG1 (low proliferative) by the Genomic Grade had a 10-year DMFS of 99%, compared with 85% for patients with GG3 (high proliferative) [61]. Thus, this molecular assessment showed to accurately define prognosis in this large proportion of patients with an intermediate nuclear grade. Regarding Ki67 positivity, it is a proven prognostic factor both in LN-negative and LN-positive early BC patients, being associated with worse DFS and OS [62]. However, its assessment is still highly variable among pathologists and no standard cut-point has been defined, making it application in clinic troublesome [63]. This was recognized by the American Society of Clinical Oncology that does not endorse the use of Ki67 for adjuvant treatment decision [49]. Even so, some guidelines state that Ki67 may be used to distinguish between Luminal A-like and Luminal B-like subtypes and recommend the use of adjuvant chemotherapy in patients with ER-positive tumors and a high Ki67 [64,65]. However, a MINDACT substudy comparing the classification of tumors by a central pathology lab using modern immunohistochemistry according to the 2013 St Gallen guidelines [66] to a molecular type classification based on Mammaprint and BluePrint™ showed provocative results [67]. In this analysis of 5806 tumor samples, 34% of tumors were classified

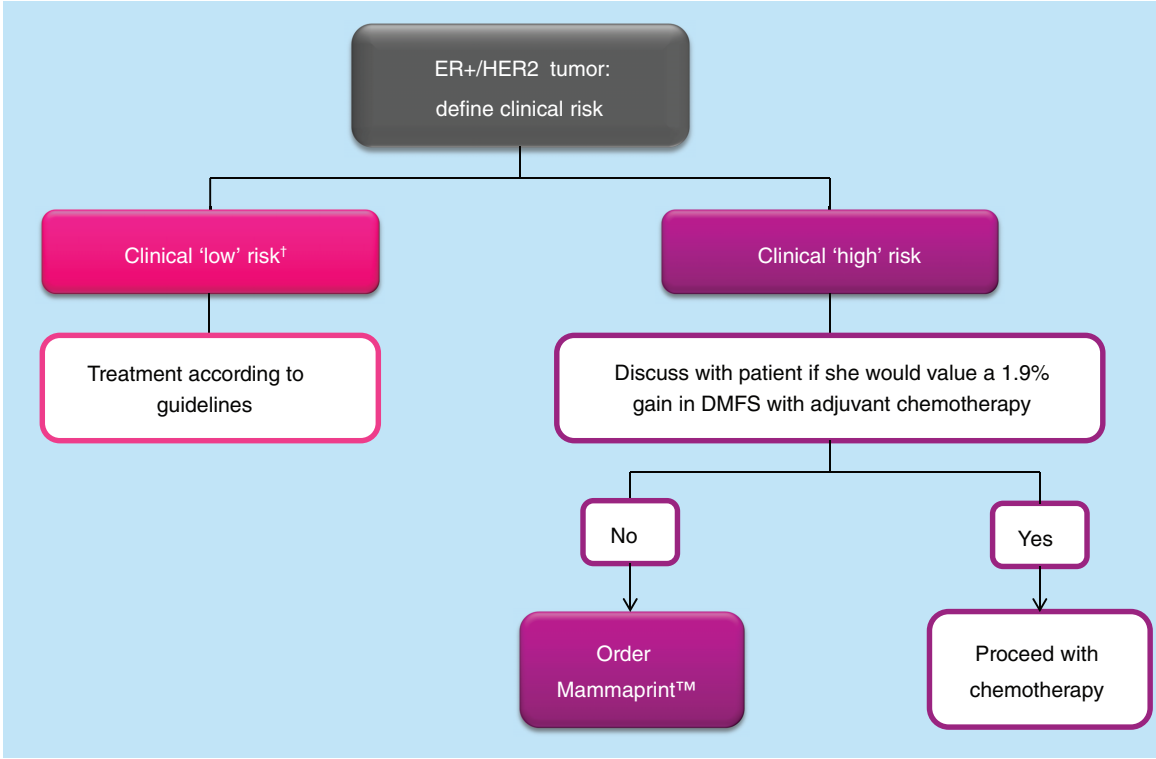

Figure 4. Practical use of Mammaprint™ in the clinic for patients with estrogen receptor-positive/HER2-negative tumors, by immunohistochemistry. With this approach, around half of the patients for which Mammaprint is ordered will be classified as low-genomic risk and could be spared chemotherapy.
†The definition of low-clinical risk used in the MINDACT trial is depicted in Figure 3.
DMFS: Distant metastasis-free survival.

Table 2. How to take Ki67 into account for the decision to order a Mammaprint™ test?

1. Assess histological grade and Ki67 score in ER-positive tumors	
• Grade & Ki67 seem to be 'concordant' – Grade 1 & Ki67 ≤10% – Grade 3 & Ki67 ≥30%	• Grade and Ki67 seem to be 'discordant' – Grade 1 & Ki67 >10% – Grade 3 & Ki67 <30%
	• Grade 2, any Ki67
2. Application in the clinic	
Consider as 'reliable information' on tumor biology and proceed as outlined in Figure 4.	Consider as 'unreliable information' → order Mammaprin™, unless the change in Grade would not impact risk as outlined in Figure 3.

The used cut-points for the Ki67 score are merely indicative; different cut-points can be used according to local pathology guidelines and practice.
ER: Estrogen receptor.

as Luminal B-like, but when applying the molecular subtyping, half of these tumors were reclassified as Luminal A (for which chemotherapy is usually not recommended). However, both classifications performed very well in predicting a DMFS rate ≥96% for the Luminal A tumors: the only difference is that the molecular classification leads to a 16% increase in tumors classified as 'Luminal A.' In the future, automated scoring of Ki67 may overcome the reproducibility issue and lead to the integration of Ki67 score in models that predict risk of BC relapse [68]. Our proposal would be to combine information regarding nuclear grade and Ki67 score, ordering a Mammaprint test for grade 2 tumors or in cases in which grade and Ki67 seem to be discordant (Table 2).

Cost–effectiveness of Mammaprint

Despite its utility, Mammaprint carries logistic and cost issues. Even if it can now be performed in formalin-fixed paraffin-embedded tissue, it is still a centralized test to be performed by Agendia, meaning that hospitals have to ship their samples to The Netherlands or to the USA, increasing the turnaround time for obtaining results. The

company is currently working to decentralize the test, creating a limited number of accredited labs across Europe. There is also an added cost to the regular pathological assessment, as the performance of the test costs, by itself, 2675€ in France [69], 3000€ in Germany [70], £2207 in the UK [71] and US$4200 in the USA [72].

Still, its utility in clinical practice was demonstrated in a pooled analysis in which the rate of prescription of chemotherapy among 790 patients decreased from 52 to 43% after performing Mammaprint [73]. Likewise, in an analysis involving more than 2000 patients, the use of Mammaprint was associated with a 10% decrease in the use of chemotherapy [74]. In the prospective WSG PRIME study [75], however, among 452 patients, there was a net increase in the use of chemotherapy of 1.9%, due to the fact that patients with low-clinical risk who were initially advised not to receive chemotherapy, were then advised to receive chemotherapy based on a high-genomic risk result, which is a strategy not proven to be beneficial in the MINDACT trial. Therefore, if this study was conducted today, probably these patients would not receive chemotherapy and the use of Mammaprint would then be associated with a reduction of 13% in the chemotherapy prescription rate. Currently, the IMPACt prospective registry (NCT02670577) is assessing the influence of Mammaprint on chemotherapy plus ET versus ET alone treatment decisions in 481 ER-positive/HER2-negative patients.

Analyses have usually shown that Mammaprint is a cost-effective test [73,76]. In the UK, the use of Mammaprint compared with standard care leads to a gain of 0.18 quality-adjusted life years (QALYs), at a cost of £1097/QALY gained [71]. In an US analysis, Oncotype DX and Mammaprint were directly compared and the results showed that Mammaprint is the most cost-effective test [77]. Another US analysis compared Mammaprint to AOL: its use spared 10% of patients from chemotherapy, with an increase of 0.15 QALYs, at a cost of US$700 to US$10,000/QALY, depending on the model used, but being cost-effective in every scenario [78]. In Spain, the use of Mammaprint compared with AOL or Oncotype DX leads to a gain in QALYs between 0.23 and 0.75, with a cost of €1457/QALY gained [79]. In contrast, a French analysis compared Mammaprint with AOL and with a chemotherapy-for-all strategy and concluded that the use of AOL would be cost-effective, but the use of Mammaprint or of chemotherapy-for-all would not [69]. However, it is important to mention that certain specificities of the French healthcare system and practice reduce the relevance of these results for other countries [72,80].

These analyses were mostly based on retrospective series and applied Mammaprint to a nonselected population. This 'blind' approach spares chemotherapy in about only a fifth of patients, corresponding to the high-clinical risk/low-genomic risk population. However, if these analyses applied the stepwise procedure recommended in the MINDACT trial of first assessing clinical risk and then ordering Mammaprint only to the high-clinical risk patients, the test would be requested to just half of the potential patients and would spare chemotherapy in half of these tested patients (the ones classified as high-clinical/low-genomic risk). Therefore, this stepwise approach would probably even further improve Mammaprint's cost–effectiveness. By contrast, in TAILORx, Oncotype DX was applied to every patient who had an indication to chemotherapy according to the National Comprehensive Cancer Network guidelines. Even after this preselection, 70% of the patients presented a low-clinical risk according to the MINDACT definition and, thus, could have been spared chemotherapy even before being tested with Oncotype DX; this suggests that for a majority of patients' traditional criteria can still determine management in the adjuvant setting [54].

Comparison with other GES

A large number of additional GES are currently available for use in clinical practice. These GES used different methods for assessing gene expression, as well as different sets of genes, generating the possibility of different performances. Though currently no data points toward better clinical outcomes for patients with one particular GES over another, several studies have compared the classification of patients according to risk by different GES with Mammaprint, when applied to the same set of samples. Table 3 summarizes the principal characteristics of the most widely used GES.

An early attempt to compare GES was made by Fan *et al.*, who tested five different GES (PAM50, Mammaprint, Oncotype DX, the 2-gene ratio component of the Breast Cancer Index and wound response) on 295 samples. Results show all tests except the 2-gene ratio predicted outcomes similarly, with a high correlation of results between patients, with concordance between Oncotype DX and Mammaprint being as high as 81% [81]. However, this was an *in silico* comparison and it was not based on the commercial GES assays, which could in part explain why these results contrast with more recent studies such as the larger OPTIMA prelim trial.

In OPTIMA prelim, 313 patients with ER-positive/HER2-negative early BC with tumors ≥3 cm and/or 1–9 positive LN were randomized between standard (chemotherapy + ET) or test-directed treatment (chemotherapy

Table 3. Available gene-expression signatures and characteristics.

Characteristics	Gene-expression signatures					
	Mammaprint™	Oncotype DX®	Prosigna® (PAM50)	Endopredict®	Breast Cancer Index	Genomic Grade
Number of genes	70	21	50	11	7	97
Method	DNA-microarray	RT-PCR	Nanostring	RT-PCR	RT-PCR	RT-PCR
Sample type	Frozen/FFPE	FFPE	FFPE	FFPE	FFPE	FFPE
Location	Central†	Central	Local	Local	Central	Central
Test results	High or low risk + subtype	High/intermediate/low risk	High/intermediate/low risk + subtype	High or low risk	High or low risk/high or low benefit	High or low risk
Clinical indication by EGTM	Predicting prognosis and guiding decision-making regarding chemotherapy for women with $ER^+/HER2^-$ EBC with LN^- or LN^+ (1–3)	Predicting prognosis and guiding decision-making regarding chemotherapy for women with $ER^+/HER2^-$ EBC with LN^- or LN^+ (1–3)	Predicting prognosis and guiding decision-making regarding chemotherapy for women with $ER^+/HER2^-$ EBC with LN^- or LN^+ (1–3)	Predicting prognosis and guiding decision-making regarding chemotherapy for women with $ER^+/HER2^-$ EBC with LN^- or LN^+ (1–3)	Predicting prognosis and guiding decision-making regarding chemotherapy for women with $ER^+/HER2^-$ EBC with LN^-	No recommendation
Prospective validation	MINDACT (positive)	TAILORx (for N0; positive)/RxPONDER (for N1; ongoing)	OPTIMA trial (ongoing)	None	None	ASTER70 (ongoing)
Agency approval	EMA, FDA	EMA, FDA	EMA, FDA	EMA, FDA	Not approved	Not approved
Original validation set	Developed in young patients (<55 years), who had not received systemic therapy following surgery	Developed in patients who had received tamoxifen only in the NSABP B-20 and B-14 studies	Postmenopausal patients in the training and develoment sets received heterogeneous treatment	Developed in postmenopausal patients, who had received ET only in the ABCSG-6 and 8 trials	Initially developed in patients treated with tamoxifen only and then further refined in heterogenously treated patients	Developed in a heterogeneous patient population, 62% aged ≤50 years, in order to permit the measurement of histologic grade via gene-expression profile
Guidelines/scientific societies that recommend the test	ASCO, EGTM, ESMO, St Gallen Panel	ASCO, EGTM, ESMO, NCCN, St Gallen Panel	ASCO, EGTM, ESMO, St Gallen Panel	ASCO, EGTM, ESMO, St Gallen Panel	ASCO, EGTM, St Gallen Panel	No recommendation

†Descentralization is ongoing to allow for local testing.
ASCO: American Society of Clinical Oncology; EBC: Early breast cancer; EGTM: European Group on Tumour Markers; ER: Estrogen receptor; ESMO: European Society for Medical Oncology; ET: Endocrine therapy; FFPE: Formalin-fixed paraffin-embedded tissue; LN: lymph node; N0: Lymph node-negative; N1: 1–3 positive lymph nodes; NCCN: National Comprehensive Cancer Network; RT-PCR: Reverse transcription polymerase chain reaction.

only if Oncotype DX recurrence score >25), with stratification by a number of other GES, including Mammaprint and Prosigna® (PAM50). Results show a significant degree of discordance between different GES, with 61% of samples having no consensus across all performed tests. Mammaprint categorized 40% patients as high-risk, while Oncotype DX only classified 18% as high-risk [71,82]. In another study, comparing Mammaprint to the 76-gene signature and Genomic Grade in 198 patients, 68% of patients were classified similarly and all three signatures added information to classic criteria and performed similarly in predicting clinical outcomes [83]. Yet, there were around 30% of patients who were assigned to different risk groups according to the test used, potentially leading to different treatment strategies, which raise important questions regarding the use of tests that have not been validated in prospective clinical trials. It is also important to note that three other large retrospective studies comparing GES, but not including Mammaprint, have concluded that Prosigna better predicted patient prognosis, including, most significantly, in terms of long-term follow-up [84–86]. The most recent of these was conducted as a preplanned comparative study in the TransATAC trial [40]. Tissue samples from women with ER-positive/HER2-negative either LN-negative (n = 591) or LN-positive with a maximum of three positives nodes (n = 183) were submitted to testing with six different prognostic prediction tools – four GES (Breast Cancer Index, Endopredict®, Prosigna, Oncotype DX) and two scores based on clinical, pathologic and imunohistochemical data (the clinical treatment score and the 4-marker immunohistochemical score). The long median follow-up available for the sample (over ten years) and mature survival data allowed for a comparative analysis of the performance of all six tools in both LN-negative and LN-positive cohorts in the 0–5 years and, critically, the 5–10-year follow-up (in which approximately 50% of the relapses occurred). The results of this study confirm that all six tools provided accurate prognostic information for LN-negative patients, especially for early relapses (<5 years). For late relapses, Breast cancer Index, Prosigna and Endopredict performed better. For LN-positive patients, however, tools integrating tumor burden

(Breast Cancer Index® and Endopredict) performed better, but the overall performance was less striking than in the node-negative subgroup. These data underscore the added value of combining both traditional criteria and GES, showing that, ideally, when determining the type and duration of adjuvant treatment, both clinical and genetic tumor factors should be taken into account to perform a comprehensive assessment of the patient's risk of relapse, leading to a truly tailored treatment.

Future perspective

Over the last 15 years, Mammaprint has proved to be a powerful prognostic tool for early BC patients, especially for ER-positive/HER2-negative with up to three positive LN. However, the first validation series showed that its ability for predicting relapse may drop after the first five years from diagnosis [12], leading to the question if this is an useful tool for the selection of patients to adjuvant treatments other than chemotherapy. Indeed, as we are still waiting for the long-term data from the MINDACT trial, an important open research question is if Mammaprint can be used for selecting the best duration of adjuvant ET. A positive hint came from the retrospective analysis of a Swedish study, in which patients with ultralow-risk tumors as defined by Mammaprint had an extremely good survival even without adjuvant ET [21]. If these findings are validated in MINDACT, this could lead to the use of Mammaprint to select patients for extended adjuvant ET.

Earlier validation studies included patients with all types of BC. However, our understanding of BC biology has evolved significantly over these past years and it is now known that the different subtypes of BC have very distinct biological and clinical behaviors [7]. Indeed, the above-mentioned meta-analysis [39,59] have shown that GES mostly based on proliferation markers are probably not useful in the triple-negative or HER2-positive group of patients as they usually have high rates of proliferation. Therefore, Mammaprint as a predictive tool for adjuvant systematic treatment should be probably used only for the ER-positive/HER2-negative subgroup of patients with high-clinical risk, as it was demonstrated in the MINDACT trial that patients with tumors with low-proliferation markers (nuclear grade 1) are usually low-clinical risk and therefore do not benefit from the use of Mammaprint [10].

On the other hand, for the neoadjuvant setting, Mammaprint is the only GES that has some data suggesting a role as predictor of benefit with neoadjuvant treatment, including in high-proliferating tumors, both with chemotherapy [24–26,45] and with the poly(ADP-ribose) polymerase (PARP) inhibitor veliparib [44], especially when a higher cut-point is used to separate high-risk from ultrahigh-risk patients. So, in the future, it is expected that Mammaprint will be increasingly used as a selection/stratifying factor in neoadjuvant trials with ER-positive/HER2-negative and/or triple-negative patients; it may also be combined with other biomarkers, like DNA-repair deficiency biomarkers, to increase its predictive ability [87].

The IMPAKT Working Group 2012 recommended the creation of registries for patients who are subjected to genomic testing in the daily clinical practice [88]. An example is the observational 'NBRST' registry (NCT01479101), including patients receiving neoadjuvant treatment after having been tested by BluePrint™/Mammaprint. It confirmed that tumors with a high-genomic risk were more chemosensitive than low-genomic risk tumors in a real-world setting [26]. However, similar registries would also be useful in other regions of the world, like Asia, in which the prognostic and clinical utility of Mammaprint has not been proven yet.

Meanwhile, new genomic profiles using DNA microarray platforms or whole genome arrays will probably be developed to explore other markers that could lead to different types of prediction (of response to treatment or of early or late relapse) in a single assay, which could be added to the Mammaprint genomic-risk [89]. Another option would be to combine a baseline tissue-based Mammaprint genomic-risk with the analysis of baseline and postneoadjuvant treatment circulating tumor-DNA to identify patients at higher risk of early relapse [90].

In conclusion, the clinical utility of Mammaprint in early BC patients has been prospectively validated by the MINDACT trial, which showed that patients with a high-clinical risk but with a low-genomic risk could be safely spared chemotherapy. Long-term survival data from MINDACT are awaited. Nevertheless, in the era of precision medicine, Mammaprint has already proven to be a useful tool for the tailoring of adjuvant chemotherapy, potentially sparing up to one in five patients from chemotherapy, with an expected decrease in toxicity and improvement of QoL.

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Executive summary

- How was the gene-expression signature (GES) developed?
  - GES were developed to define which early breast cancer patients benefit from the use of adjuvant chemotherapy, as it leads to important short- and long-term consequences for both patient and society.
  - Mammaprint™ was developed in 2002 as a 70-gene signature with the ability of distinguishing patients with high risk of distant metastasis at 5 years ('high-genomic risk') from those with a low risk ('low-genomic risk').
- Problems and controversies regarding the test and the MINDACT trial.
  - The MINDACT trial was not powered to directly compare outcomes between patients receiving chemotherapy versus no-chemotherapy, therefore the predictive ability of Mammaprint for selecting patients who would benefit the most from chemotherapy could not be proven.
  - Its prognostic ability in the adjuvant setting for HER-positive and triple-negative patients has been questioned, as most of these patients are classified as 'high-genomic risk'.
- Practical application in the clinic.
  - Mammaprint is the first GES to have its clinical utility validated in a prospective randomized trial (the MINDACT trial) as a decision-making tool when genomic-risk results are discordant with clinical criteria.
  - Mammaprint is approved by many health authorities throughout the world, based on the results of extensive studies on its analytical validity, clinical validity, and more recently, clinical utility in a prospective setting (the MINDACT trial).
  - Mammaprint was incorporated in the 8th TNM edition for the classification of breast cancer, along other GES, as a way of downstaging selected estrogen receptor-positive/HER2-negative node-negative tumors with low-genomic risk into the same prognostic category as T1a-T1bN0M0.
- Use of Mammaprint in clinical trials.
  - In the neoadjuvant setting, a higher cut-point separates patients with high-risk from ultrahigh-risk, which predicts the sensitivity to chemotherapy, even in high proliferation tumors like HER2-positive and triple-negative and lead to the use of Mammaprint as a stratification factor in the neoadjuvant I-SPY2 trial.
- Cost–effectiveness of Mammaprint.
  - Cost–effectiveness analyses have mostly shown that Mammaprint is likely to be a cost-effective test.
- Precision medicine.
  - In the era of precision medicine, Mammaprint has already proven to be an important tool for the tailoring of adjuvant chemotherapy, potentially sparing an important proportion of patients from aggressive treatments and leading to a reduction in toxicities and costs, with an expected improvement in the quality of life of early breast cancer patients.

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